ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

FIRST YEAR (COMMON TO ALL BRANCHES) — SEMESTER 2

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Classify Engineering Materials (Ferrous, Non-ferrous, Polymers, Ceramics). Describe Iron-Carbon equilibrium diagram. [14 Marks]

[OR]

Q.1 (Alt) Explain Mechanical Properties: Tensile strength, Hardness, Toughness, Ductility, Malleability. [14 Marks]

--- UNIT II ---

Q.2 Explain Working of 2-Stroke and 4-Stroke Petrol (SI) and Diesel (CI) Engines with P-v diagrams. [14 Marks]

[OR]

Q.2 (Alt) Compare SI vs CI engines in terms of compression ratio, fuel injection, and thermal efficiency. [14 Marks]

--- UNIT III ---

Q.3 Explain Steam Boilers: Classify Fire-Tube vs Water-Tube boilers. Describe Cochran and Babcock-Wilcox boilers. [14 Marks]

[OR]

Q.3 (Alt) Explain Boiler Mountings (Safety valve, Water level indicator) and Accessories (Superheater, Economizer). [14 Marks]

--- UNIT IV ---

Q.4 Explain Power Transmission drives: Belt drives (Open, Cross), Rope drives, Chain drives, Gear trains. [14 Marks]

[OR]

Q.4 (Alt) Derive Ratio of Tensions for flat belt drive (T1/T2 = e^(mu theta)). Explain Velocity Ratio. [14 Marks]

--- UNIT V ---

Q.5 Describe Lathe Machine operations: Turning, Facing, Thread cutting, Taper turning, Knurling. [14 Marks]

[OR]

Q.5 (Alt) Explain Drilling and Milling machine operations. Describe basic refrigeration vapor compression cycle. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****